

Practice Problems 9

A transparent film has a thickness of 0.003250 cm, and a refractive index of 1.4000. Find (a) the order of interference m at $\theta = 0^\circ$ and (b) the first four angles at which red light of wavelength 6500 Å will form bright-light fringes.

Ans. (a) $m = 100$, (b) 5.73, 9.94, 12.84, and 15.20°

A thin film has a thickness of 0.04650 cm and a refractive index of 1.5230. Find the angle ϕ at which the dark fringe 122.5 will be observed if monochromatic light 6560 Å is used as an extended source.

In an experiment involving Newton's rings, the diameters of the fifth and fifteenth bright rings formed by sodium yellow light are measured to be 2.303 and 4.134 mm, respectively. Calculate the radius of curvature of the convex glass surface.

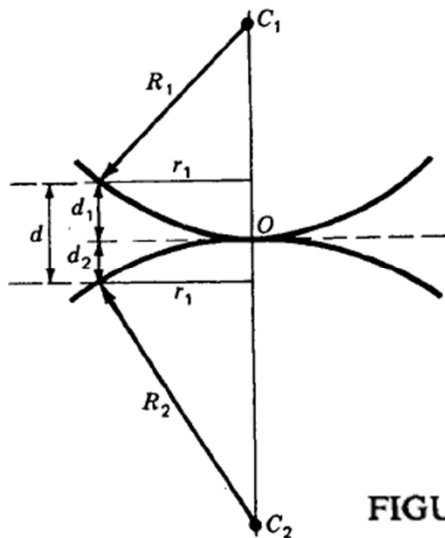


FIGURE P14.4

Three spherical glass surfaces of unknown radius are brought into contact in pairs, and each pair used to form Newton's rings. The diameters of the twenty-fifth bright fringe for the three possible combinations are 8.696, 9.444, and 10.2680 mm, respectively. Find (a) the path difference d and (b) the radii of the three glass surfaces. Assume the mercury green light used has a wavelength of 5461 Å. See Fig. P14.4.